

第一章

1. 如表

近似值	有效数字位数	误差限	相对误差限	
123.45	5	$\frac{1}{2} \times 10^{-2}$	4.05×10^{-5}	5×10^{-5}
0.83500	5	$\frac{1}{2} \times 10^{-5}$	5.99×10^{-6}	6.25×10^{-6}
0.015	2	$\frac{1}{2} \times 10^{-3}$	3.33×10^{-2}	5×10^{-2}
10.008	5	$\frac{1}{2} \times 10^{-3}$	4.99×10^{-5}	5×10^{-5}

3. $\frac{1}{2} \times 10^{-2}$

4. $\epsilon(s(t^*)) = 0.1 \times gt^*, \epsilon_r(s(t^*)) = 0.2 \times \frac{1}{t^*}$

5. 如表

函数	误差限	相对误差限
$x + y$	5.5×10^{-4}	4.37×10^{-4}
$x - y$	5.5×10^{-4}	7.32×10^{-4}
xy	5.15×10^{-4}	2.03×10^{-3}
$\frac{x}{y}$	8.05×10^{-3}	2.03×10^{-3}

第二章

1. $p_2(x) = N_2(x) = -0.6250x^2 + 2.0125x - 0.6238, R(1.75) \leq 1.038 \times 10^{-7},$
 $p_2(1.75) = N_2(1.75) = 0.9840$

3. $p_3(x) = N_3(x) = -0.1167x^3 + 1.6500x^2 - 7.1333x + 9.6$

6. $p(2.5) = 4.5$

7. $H_3(x) = -2x^3 + 8x^2 - 9x + 5$

8. $p_4(x) = 0.25x^4 - 1.5x^3 + 2.25x^2$

第三章

1. $x_1 = 0.7143, x_2 = -0.0714$

2. $y = a + bx^2, a = 0.9726, b = 0.0500$

3. $y = \frac{1}{a+bx}, a = -2.0536, b = 3.0267$

4. $y = a\theta + b\theta^2, a = 0.9497, b = -0.1129$

第四章

1. $S_T = 2.1836, S_S = 2.0263, e_T \leq 0.6796, e_S \leq 0.0689$

2. $S = 17.3264$

5. $S_T = 1.97423, n = 2; S_S = 2.00026, n = 4$

6. $n = 6$

9. $\int_0^1 f(x)dx \approx \frac{1}{2}f\left(\frac{1}{2} - \frac{1}{2\sqrt{3}}\right) + \frac{1}{2}f\left(\frac{1}{2} + \frac{1}{2\sqrt{3}}\right)$

第五章

1. $x_1 = -4, x_2 = 1, x_3 = 2$

2. $x_1 = 109, x_2 = 27, x_3 = -66$

3. $x_1 = 1, x_2 = \frac{1}{2}, x_3 = 1$

第六章

2. Jacobi: $X^{k+1} = \begin{bmatrix} 0 & \frac{1}{5} & \frac{1}{5} \\ \frac{1}{5} & 0 & \frac{1}{10} \\ \frac{1}{3} & \frac{2}{3} & 0 \end{bmatrix} X^k + \begin{bmatrix} \frac{1}{10} \\ \frac{1}{20} \\ \frac{1}{3} \end{bmatrix}$, 收敛; Seidel: $X^{k+1} = \begin{bmatrix} 0 & \frac{1}{5} & \frac{1}{5} \\ 0 & \frac{1}{25} & \frac{7}{50} \\ 0 & \frac{7}{75} & \frac{4}{25} \end{bmatrix} X^k + \begin{bmatrix} \frac{1}{10} \\ \frac{7}{100} \\ \frac{31}{75} \end{bmatrix}$, 收敛

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3. 如下

A 矩阵: $\lambda_1 = \lambda_2 = \lambda_3 = 0$, Jacobi 收敛; $\lambda_1 = 0, \lambda_2 = \lambda_3 = 2$, Seidel 不收敛

B 矩阵: $\lambda_1 = 0, \lambda_2 = \frac{\sqrt{5}}{2}i, \lambda_3 = -\frac{\sqrt{5}}{2}i$, Jacobi 不收敛; $\lambda_1 = 0, \lambda_2 = \lambda_3 = -\frac{1}{2}$, Seidel 收敛

第七章

1. $k = 7, x_7 = -1.792969$

4. $|\phi'_1(x)| > 1$, 不收敛

9. $x_{k+1} = \frac{2x_k^3+1}{3x_k^2-1}, x^* = 1.324718$

11. $-\frac{2}{f'(x)} < c < 0$ ($f'(a) > 0$); $0 < c < -\frac{2}{f'(a)}$ ($f'(a) < 0$)